

Некоторые квантово-механические свойства модели Вольфрама

Джонатан Горард • Опубликовано в [Complex Systems](#) 25 июня 2020 • Физика

TLDR В этой статье доказывается (с использованием различных комбинаторных и теоретико-порядковых методов), что геометрия многопутевого эволюционного графа в пределе континуума сходится к геометрии комплексного проективного гильбертова пространства. На основе этой информации мы выводим аналог уравнений поля Эйнштейна для всей многопутевой системы.

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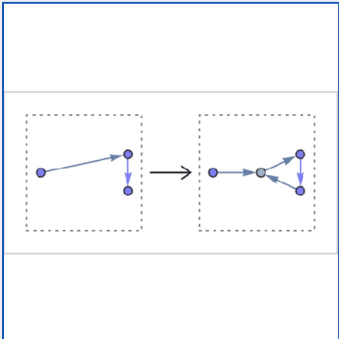


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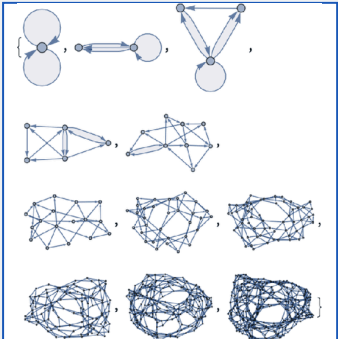


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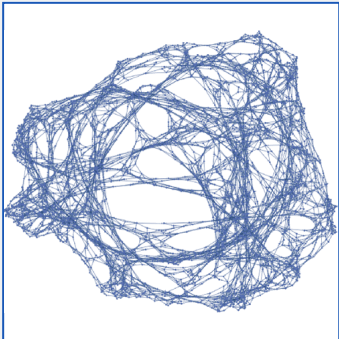


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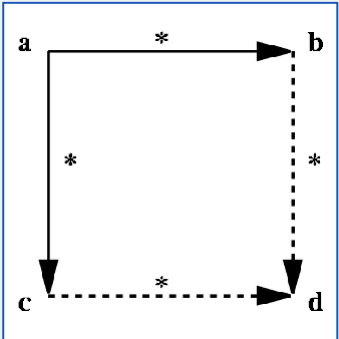


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Некоторые релятивистские и гравитационные свойства модели Вольфрама

Джонатан Горард

Физика · [Сложные системы](#)

2020

TLDR Доказано, что каузальная инвариантность (а именно требование, чтобы все каузальные графы были изоморфны, независимо от выбора порядка обновления гиперграфа) эквивалентна дискретной версии общей ковариантности, при этом изменения в порядке обновления соответствуют дискретным калибровочным преобразованиям, и выводится дискретный аналог лоренц-ковариантности. [Развернуть](#)

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Аксиоматическая квантовая теория поля в дискретном пространстве-времени через многопутевую причинно-следственную структуру: случай энтропии запутанности

Джонатан Горард Джулия Даннеманн-Фрайтаг

Физика · [arXiv.org](#)

2023

TLDR Показано, как можно строго построить аксиоматическую теорию поля для свободного безмассового скалярного поля (подчиняющегося соответствующим бозонным коммутационным соотношениям) на основе многопутевых каузальных графов — комбинаторной структуры, достаточно общей, чтобы включать в себя в качестве частных случаев как каузальные множества, так и эволюцию моделей Вольфрама. [Развернуть](#)

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Emergent gravity from the correlation of spin- $\frac{1}{2}$ systems coupled with a scalar field

Quentin Ansel

Physics
2024

This paper introduces several ideas of emergent gravity, which come from a system similar to an ensemble of quantum spin- $\frac{1}{2}$ particles. To derive a physically relevant theory, the model is... [Expand](#)

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Quantum Gates from Wolfram Model Multiway Rewriting Systems

Furkan Semih Dündar X. Arsiwalla Hatem Elshatlawy

Physics, Computer Science • [arXiv.org](#)
2025

TLDR It is shown how representations of finite-dimensional quantum operators can be constructed using nondeterministic rewriting systems, and found that, as formal models of nondeterministic computation, rewriting systems of Leibnizian strings with causal structure encode representations of the CNOT, $\pi/8$, and Hadamard gates.[Expand](#)

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ZX-Calculus and Extended Hypergraph Rewriting Systems I: A Multiway Approach to Categorical Quantum Information Theory

Jonathan Gorard Manojna Namuduri X. Arsiwalla

Computer Science, Physics · [arXiv.org](#)

2020

TLDR The objective of the present article is to begin elucidating the formal correspondence between the two methodologies in the context of the ZX-calculus formalism of Coecke and Duncan for reasoning diagrammatically about linear maps between qubits. [Expand](#)

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Algorithmic Causal Sets and the Wolfram Model

Jonathan Gorard

Physics, Mathematics · [arXiv.org](#)

2020

TLDR It is demonstrated that the hypergraph rewriting approach of the Wolfram model can effectively be interpreted as providing an underlying algorithmic dynamics for causal set evolution and causal invariance can be used to infer conformal invariance of the induced causal partial order. [Expand](#)

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Growth Functions, Rates and Classes of String-Based Multiway Systems

Yorick Zeschke

Physics, Computer Science · [Complex Systems](#)

2022

TLDR It is proven that string rewriting systems, while never exceeding exponential functions in their growth, are capable of growing arbitrarily slowly, that is, slower than the asymptotic inverse of every Turing-computable function. [Expand](#)

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Computational General Relativity in the Wolfram Language using Gravitas I: Symbolic and Analytic Computation

Jonathan Gorard

Physics, Computer Science · [arXiv.org](#)

2023

TLDR The design and capabilities of Gravitas's symbolic subsystem, including its general and flexible handling of arbitrary geometries parametrized by arbitrary curvilinear coordinate systems, as well as its various high-level tensor calculus and differential geometry features are focused on.

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A Functorial Perspective on (Multi)computational Irreducibility

Jonathan Gorard

Computer Science, Mathematics · [arXiv.org](#)

2023

TLDR It is argued that the irreducibility of (multi)computations may be thought of as being dual to the locality of time evolution in functorial approaches to quantum mechanics and quantum field theory. [Expand](#)

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Homotopies in Multiway (Non-Deterministic) Rewriting Systems as n-Fold Categories

X. Arsiwalla Jonathan Gorard Hatem Elshatlawy

Computer Science, Mathematics · Complex Systems
2022

TLDR It is shown that a multiway rewriting system with homotopies up to order n may naturally be formalized as an n -fold category, such that (upon inclusion of appropriate inverse morphisms via invertible rewriting relations) the infinite limit of this structure yields an ∞ -groupoid. [Expand](#)

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Some Relativistic and Gravitational Properties of the Wolfram Model

Jonathan Gorard

Physics · [Complex Systems](#)

2020

TLDR It is proved that causal invariance (namely, the requirement that all causal graphs be isomorphic, irrespective of the choice of hypergraph updating order) is equivalent to a discrete version of general covariance, with changes to the updating order corresponding to discrete gauge transformations, and a discrete analog of Lorentz covariance is deduced. [Expand](#)

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Quantum speed limits: from Heisenberg's uncertainty principle to optimal quantum control

Sebastian Deffner

S. Campbell

Physics

2017

One of the most widely known building blocks of modern physics is Heisenberg's indeterminacy principle. Among the different statements of this fundamental property of the full quantum mechanical... [Expand](#)

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Decoherence, einselection, and the quantum origins of the classical

W. Zurek

Physics

2003

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Geometric quantum mechanics

D. Brody

L. Hughston

Physics

2001

Abstract The manifold of pure quantum states can be regarded as a complex projective space endowed with the unitary-invariant Fubini–Study metric. According to the principles of geometric quantum... [Expand](#)

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Institute for Mathematical Physics Geometrical Formulation of Quantum Mechanics Geometrical Formulation of Quantum Mechanics

A. Ashtekar

Troy A. Schilling

Physics, Mathematics

States of a quantum mechanical system are represented by rays in a complex Hilbert space. The space of rays has, naturally, the structure of a Kähler manifold. This leads to a geometrical...

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Zeno's paradox in quantum theory

B. Misra

E. Sudarshan

Physics

1976

A quantum-theoretic expression is sought for the probability that an unstable particle prepared initially in a well-defined state will be found to decay sometime during a given interval. It is argued... [Expand](#)

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"Relative State" Formulation of Quantum Mechanics

H. Everett

Physics

1957

The task of quantizing general relativity raises serious questions about the meaning of the present formulation and interpretation of quantum mechanics when applied to so fundamental a structure as... [Expand](#)

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On Unitary Evolution and Collapse in Quantum Mechanics

F. Giacosa

Physics

2014

In the framework of an interference setup in which only two outcomes are possible (such as in the case of a Mach–Zehnder interferometer), we discuss in a simple and pedagogical way the difference... [Expand](#)

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Quantum mechanical hamiltonian models of turing machines

P. Benioff

Physics, Computer Science
1982

Quantum mechanical Hamiltonian models, which represent an arbitrary but finite number of steps of any Turing machine computation, are constructed here on a finite lattice of spin-1/2 systems....

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The stretched horizon and black hole complementarity.

L. Susskind L. Thorlacius J. Uglum

Physics · [Physical Review D, Particles and fields](#)
1993

Three postulates asserting the validity of conventional quantum theory, semiclassical general relativity, and the statistical basis for thermodynamics are introduced as a foundation for the study of... Expand

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